1. Algebra

1. Binomial Expansion

We learnt the following expansions in G.C.E (O/L)

$(a+b)^2$, $(a-b)^2$, $(a+b)^3$ and $(a-b)^3$

Now we will obtain the above binomial expansions.

$(a+b)^2 = (a+b)(a+b)$

$= a(a+b) + b(a+b)$

$= (a^2 + ab + ab + b^2$

$= a^2 + 2ab + b^2$ ———— (1)

$(a-b)^2 = (a-b)(a-b)$

$= a(a-b) - b(a-b)$

$= a^2 - ab - ab + b^2$

$= a^2 - 2ab + b^2$ ———— (2)

OR

The result (2) above can be obtained by using the result (1).

$(a-b)^2 = [a+(-b)]^2$

Replacing b, by $(-b)$ in the result (1)

ie $(a+b)^2 = a^2 + 2ab + b^2$

$[a+(-b)]^2 = a^2 + 2a(-b) + (-b)^2$

$(a-b)^2 = a^2 - 2ab + b^2$

$(a+b)^3 = (a+b)(a+b)^2$

$= (a+b)(a^2 + 2ab + b^2)$

$= a(a^2 + 2ab + b^2) + b(a^2 + 2ab + b^2)$

$= a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3$

$= a^3 + 3a^2b + 3ab^2 + b^3$ ————(3)

$(a-b)^3 = (a-b)(a-b)^2$

$= (a-b)[a^2 - 2ab + b^2]$

$= a(a^2 - 2ab + b^2) - b(a^2 - 2ab + b^2)$

$= a^3 - 2a^2b + ab^2 - a^2b + 2ab^2 - b^3$

$= a^3 - 3a^2b + 3ab^2 - b^3$ ———— (4)

OR

The expansion of $(a - b)^3$ can be obtained by replacing b **by $(-b)$ in (3).**

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$[a + (-b)]^3 = a^3 + 3a^2(-b) + 3a(-b)^2 + (-b)^3$$

$$= a^3 - 3a^2b + 3ab^2 - b^3$$

$$(a + b + c)^2 = [(a + b) + c]^2$$

$$= (a + b)^2 + 2(a + b)c + c^2$$

$$= a^2 + 2ab + b^2 + 2ac + 2bc + c^2$$

$$= a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$(a + b)^2$	$= a^2 + 2ab + b^2$
$(a - b)^2$	$= a^2 - 2ab + b^2$
$(a + b)^3$	$= a^3 + 3a^2b + 3ab^2 + b^3$
$(a - b)^3$	$= a^3 - 3a^2b + 3ab^2 - b^3$
$(a + b + c)^2$	$= a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

We will apply the above results in solving related problems.

Example 1

Expand the following binomials.

(i) $(2x + 3y)^2$ (ii) $(2xy - 5z)^2$

(iii) $(3x + 2y)^3$ (iv) $\left(ab - \frac{2}{c}\right)^3$

(v) $(a + b - c)^2$

(i) $(2x + 3y)^2 = (2x)^2 + 2 \times 2x \times 3y + (3y)^2$

$= 4x^2 + 12xy + 9y^2$

(ii) $(2xy - 5z)^2 = (2xy)^2 - 2 \times 2xy \times 5z + 3(5z)^2$

$= 4x^2y^2 - 20xyz + 25z^2$

(iii) $(3x + 2y)^3 = (3x)^3 + 3 \times (3x)^2 \times (2y) + 3 \times (3x) \times (2x)^2 + (2y)^3$

$= 27x^3 + 54x^2y + 36xy^2 + 8y^3$

(iv) $\left(ab - \frac{2}{c}\right)^3 = (ab)^3 - 3(ab)^2 \times \frac{2}{c} + 3(ab)\left(\frac{2}{c}\right)^2 - \left(\frac{2}{c}\right)^3$

$= a^3b^3 - \frac{6^2a^2b^2}{c} + \frac{12ab}{c^2} - \frac{8}{c^3}$

(v) $(a + b - c)^2 = a^2 + b^2 + c^2 + 2ab - 2bc - 2ca$

Example 2

Given that $a + b = 4$ **and** $ab = 5$ **find the value of (i)** $a^2 + b^2$ and (ii) $a^3 + b^3$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$a^2 + b^2 = (a + b)^2 - 2ab$$

$$= 4^2 - 2 \times 5 = 16 - 10 = 6$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$a^3 + b^3 = (a + b)^3 - 3a^2b - 3ab^2$$

$$= (a + b)^3 - 3ab(a + b)$$

$$= 4^3 - 3 \times 5 \times 4$$

$$= 64 - 60 = 4$$

Exercise 1

Write the expansion of the following:

1. $(2a + 3b)^2$
2. $(3a - 4b)^2$
3. $\left(x + \frac{1}{x}\right)^2$
4. $(2xy + 5z)^2$
5. $\left(\frac{1}{a} + \frac{1}{b}\right)^2$
6. $\left(x - \frac{1}{x}\right)^2$
7. $\left(\frac{a}{2} - \frac{2}{a}\right)^2$
8. $\left(\frac{1}{a} - \frac{2}{b}\right)^2$
9. $(4xy - 3z)^2$
10. $(a + 2b)^3$
11. $(2a - b)^3$
12. $(3a + 2b)^3$
13. $\left(x + \frac{1}{x}\right)^3$
14. $\left(x - \frac{1}{x}\right)^3$
15. $(ab - 2c)^3$
16. $\left(\frac{1}{a} + \frac{1}{b}\right)^3$
17. $\left(\frac{1}{a} - \frac{2}{b}\right)^3$
18. $(2xy - 3z)^3$
19. $(a + b + c)^2$
20. $(a + b - c)^2$
21. $(a - b + c)^2$
22. $(a - b - c)^2$
23. $(a - 2b + c)^2$
24. $(a - b - 2c)^2$
25. Evaluate

    (i) 101^3 (ii) 198^3 (iii) 401^3 (iv) 999^3

26. Evaluate

    (i) $101^2 + 2 \times 101 \times 99 + 99^2$ (ii) $88^2 - 2 \times 88 \times 87 + 87^2$

27. Evaluate

(a) $51^3 + 3 \times 51^2 \times 49 + 3 \times 51 \times 49^2 + 49^3$

(b) $101^3 - 3 \times 101^2 \times 99 + 3 \times 101 \times 99^2 - 99^3$

28. Show that

(i) $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$

(ii) $(a+b)^2 - (a-b)^2 = 4ab$

(iii) $(a+b)^3 + (a-b)^3 = 2a(a^2 + 3b^2)$

(i) $(a+b)^3 - (a-b)^3 = 2b(3a^2 + b^2)$

29. If $x + \frac{1}{x} = a$, find the values of (a) $x^2 + \frac{1}{x^2}$ and (b) $x^3 + \frac{1}{x^3}$ in terms of a.

30. If $x - y = 4$ and $xy = 21$, find the value of $x^3 - y^3$

31. If $(x+y) = -\frac{1}{3}$, find the value of $x^3 + y^3 - xy$

32. If $a - \frac{1}{a} = -5$, find the value of $a^3 - \frac{1}{a^3} - 200$

33. If $\frac{x^2 - 1}{x} = 4$, show that $\frac{x^6 - 1}{x^3}$ is 76

34. If $\frac{a^2 - 1}{a} = 2$, find the value of $\frac{a^6 - 1}{a^3}$

35. If $a + b - 3 = 0$, find the value of $a^3 + b^3 + 9ab - 26$

36. If $a + b - 7 = 0$ and $ab = 12$, find the value of $a^3 + b^3 + 4ab(a+b)$

37. If $p = 2q + 4$, show that $p^3 - 8q^3 - 24pq = 64$

38. If $a + b + c = 0$, show that $a^3 + b^3 + c^3 = 3abc$

39. If $p + q = 1 + pq$, show that $p^3 + q^3 = 1 + p^3q^3$

40. If $ab(a+b) = p$, show that $a^3 + b^3 + 3p = \frac{p^3}{a^3b^3}$